جامعة حفر الباطن
University of Hafr Al Batin

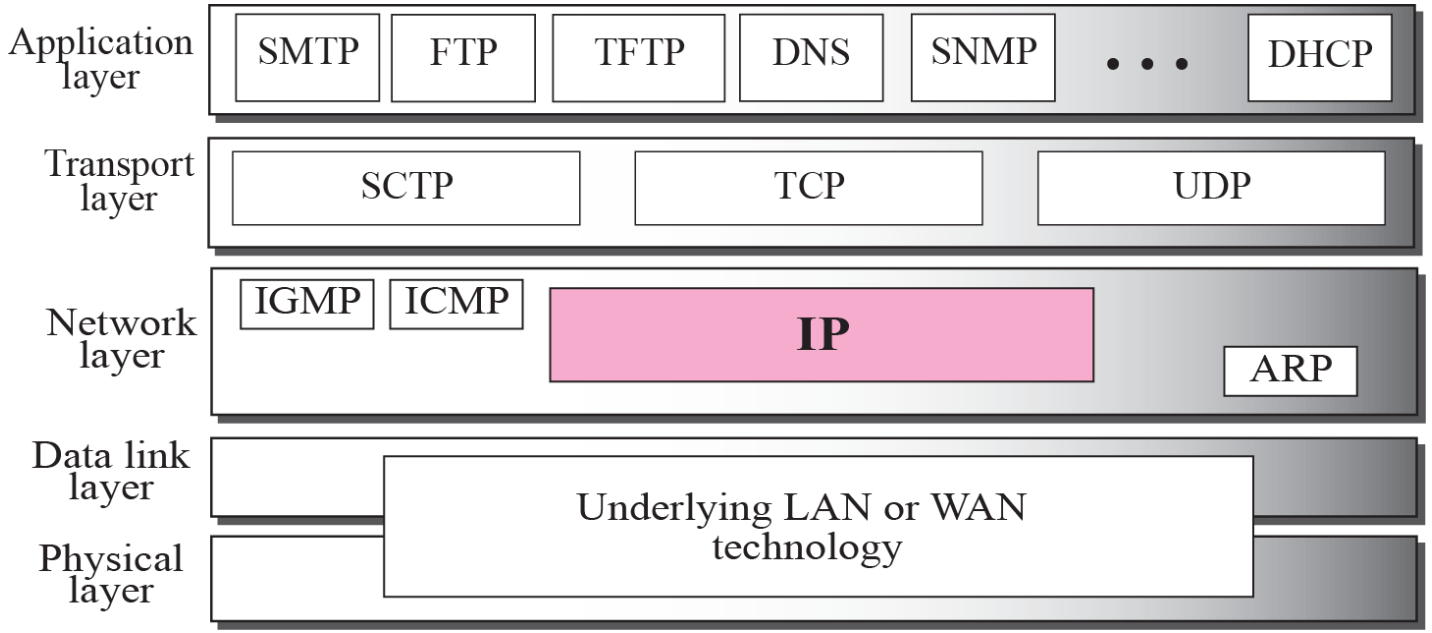
Lecture 4 - Internet Protocol Version 4 (IPv4)

Objectives

- IP protocol and the position of IP in TCP/IP protocol suite.
- IPv4 datagram format.
- Datagram fragmentation and reassembly.
- IPv4 datagram options and their applications.
- IPv4 datagram checksum.
- IPv4 Addressing

Internet Protocol (IP)

- The is the transmission mechanism used by the TCP/IP protocols at the network layer.



Datagrams

- Packets in the network (internet) layer are called **datagrams**.
- A datagram is a variable-length packet consisting of two parts:
  - Header
  - Data.
- The header is 20 to 60 bytes in length and contains information essential to routing and delivery.
- It is customary in TCP/IP to show the header in 4-byte sections.

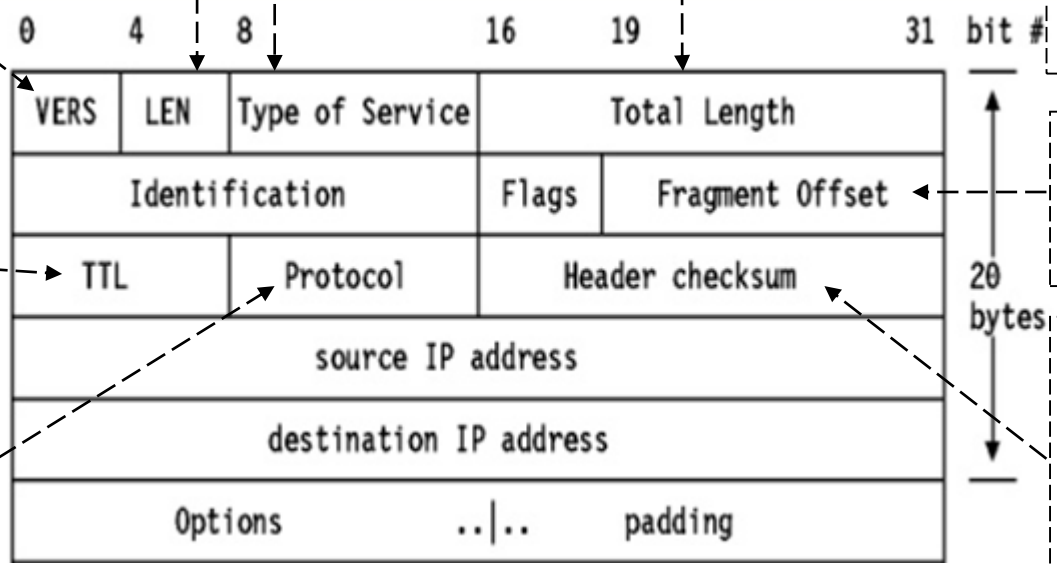
IP: datagram header

Header length (in 4 bytes)

Datagram's priority for Queuing

Size of he datagram (header + data)

IP version number



Time To Live: max number remaining hops (decremented at each router)

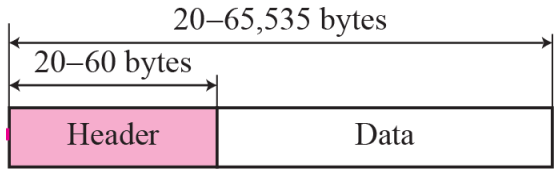
They are used for fragmentation and reassembly

Upper layer protocol to deliver payload

The router calculates the checksum of the header and compares it to the checksum field

IP Datagram Format

- The **Total Length** field defines the total length of the datagram including the header.
  - It can range from 20 to 65535 bytes.
- The **Header Length** is the length of the header in 32-bit increments.
  - If the value was 5, then the header length is 5x4=20 bytes.
  - The header length can range from 20 bytes (no options in the header) to 60 bytes (with options and padding).



a. IP datagram

IP Datagram Format

- **Question:**
  - An IP packet has arrived with the first 8 bits as: 01000010 . The receiver discards the packet. Why?

- **Solution:**
  - The 4 left-most bits (0100)show the version, which is correct (4).
  - The next 4 bits (0010) show the header length (2 × 4 = 8).
  - The minimum number of bytes in the header must be 20.
  - The packet has been **corrupted** in transmission.

IP Datagram Format

- **Question:**
  - In an IP packet, the value of HLEN is 1000 in binary. How many bytes of options are being carried by this packet?
- **Solution:**
  - The HLEN value is 8
  - Which means the total number of bytes in the header is 8 × 4 or 32 bytes.
  - The first 20 bytes are the base header
  - the next **12 bytes are the options**.

Fragmentation

- A datagram can travel through different networks.
- Each router decapsulates the IP datagram from the frame it receives, processes it, and then encapsulates it in another frame.
- The format and size of the received frame depend on the protocol used by the physical network through which the frame has just traveled.
- The format and size of the sent frame depend on the protocol used by the physical network through which the frame is going to travel.

IP Datagram Options

- The header of the IP datagram is made of two parts: a fixed part and a variable part.
  - The *fixed* part is 20 bytes long and was discussed earlier.
  - The *variable* part is the <u>options</u>, which can be a maximum of 40 bytes.

- Options are ***not*** *required* for a datagram.
  - They can be used for network testing and debugging.
  - Although options are not a required part of the IP header, option processing is required of the IP software.

IP Datagram Options

Option: Type (8 bits / 1 byte)

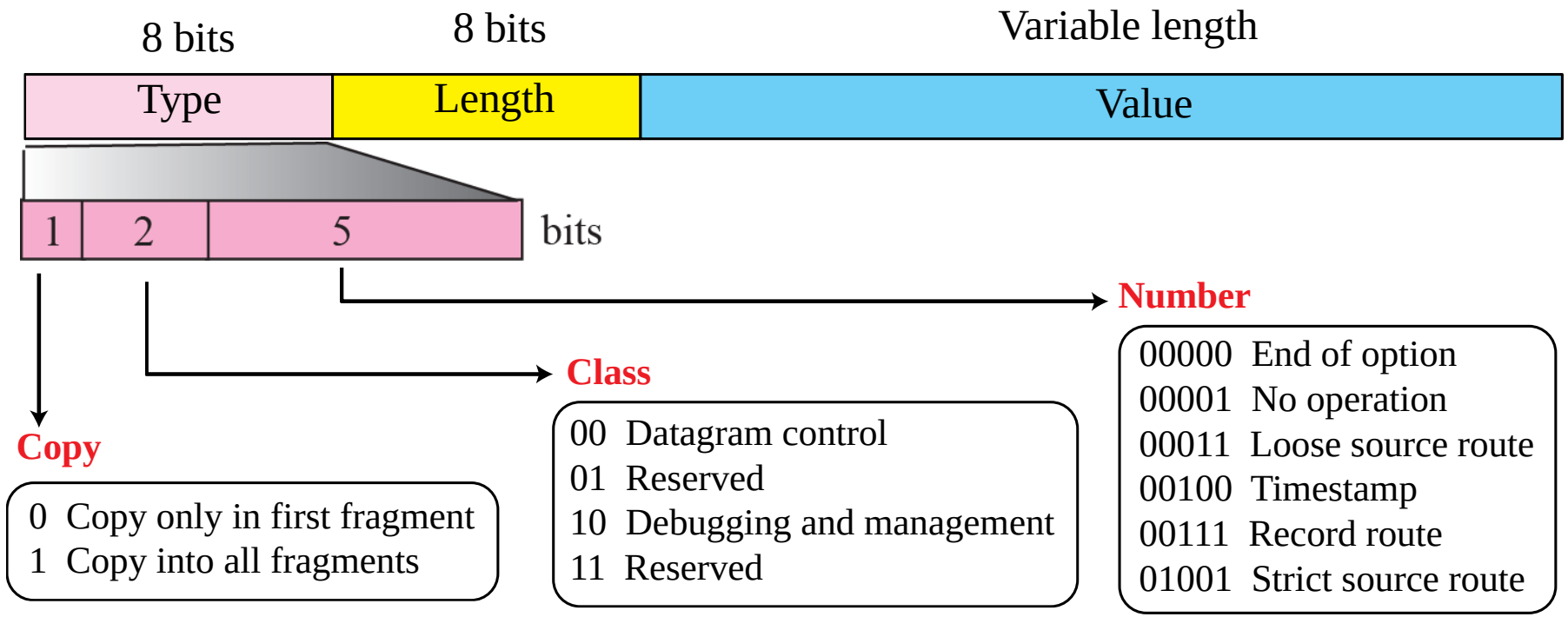
- Option Type is divided into three sub-subfields:
  - **Copy (1 bit):** this bit is set to 1 if the option is intended to be copied into all fragments. It is set to 0 if the option should *not* be copied.
  - **Class (2 bits):** defines the general category (class) of the option. We have 4 possible categories, only 2 are used.
  - **Number (5 bits):** Specifies the kind of option. You can specify 32 options for each class.

Option: Length (8 bits / 1 byte):

- The size of the entire option, including all three subfields here, *in bytes*.

IP Datagram Options

- *Option format*



IP Datagram Options

- Examples:
  1. Option starts with 00000000:
     - Copy: No.
     - Class: Datagram Control
     - Number: End of Option
     - Used to mark the end of a list of options.

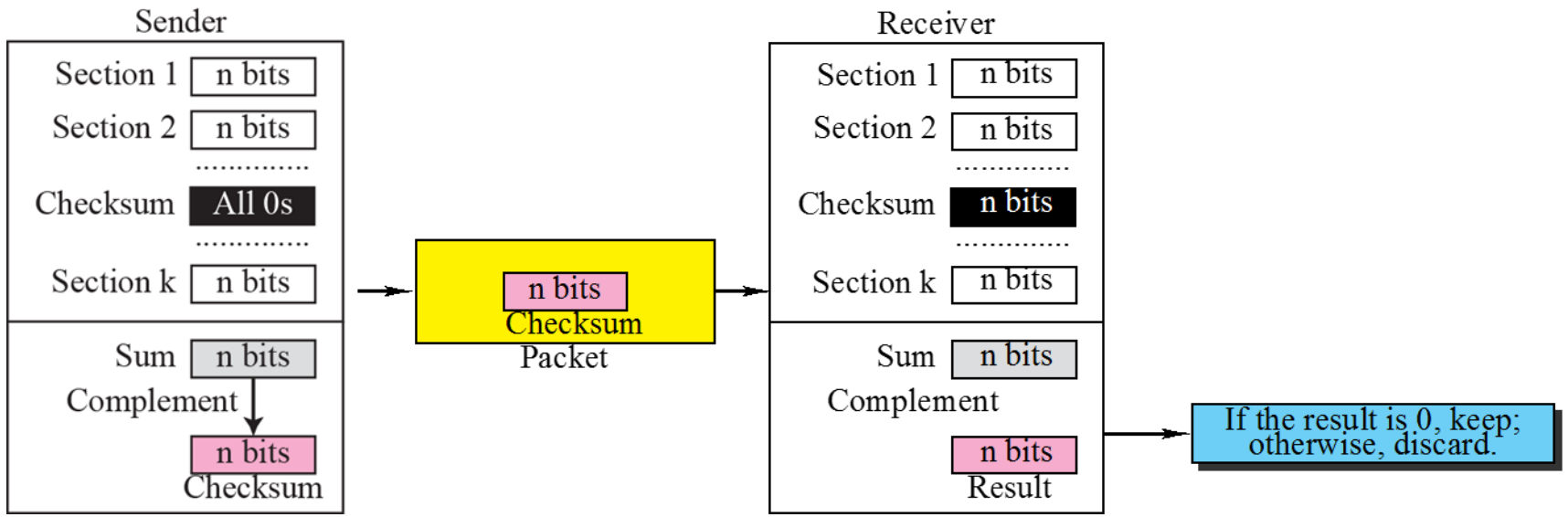
  2. Option starts with 10000001:
     - Copy: Yes. Copy the option into all fragments.
     - Class: Datagram Control
     - Number: No operation
     - used as “internal padding” to align certain options on a 32-bit boundary when required

Checksum

- The error detection method used by most TCP/IP protocols is called the checksum.
- The checksum protects against the corruption that may occur during the transmission of a packet.
- It is redundant information added to the packet.
- How the Checksum is done?
  1. The checksum is calculated at the sender and the value obtained is sent with the packet.
  2. The receiver repeats the same calculation on the whole packet including the checksum.
  3. If the result is zero, the packet is accepted; otherwise, it is rejected.

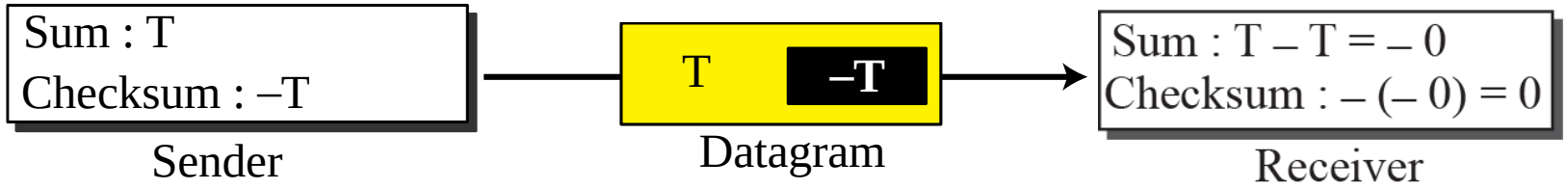
Checksum

- Checksum concept:
  - Checksum in IP covers only the header, not the data.



Checksum

- Checksum arithmetic:



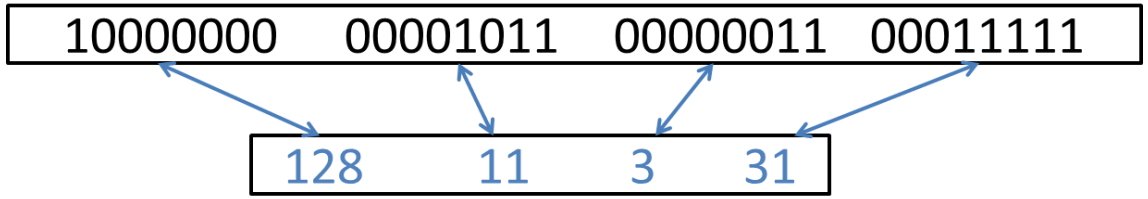
IP Addressing

- The IP address is the identifier used in the Network layer of the TCP/IP protocol to identify each device connected to the Internet.
- IP address is unique and provide a network identity for the node.
- IPv4 is a 32-bit address that uniquely and universally defines a host or router connection to the Internet.
- IP address is the address of the connection and not the host or router address. A change in location may change the IP address

IP Addressing

- **Address Space:** is the total number of addresses used by the protocol.
- If a protocol uses b bits to define an address, the address space is 2^b.
  - IPv4 uses 32-bit address, hence the address space is 2^{32} or 4,294,967,296.

- IPv4 address is divided into four fields called octets.
  - Each octet represent 8 bits of the 32-bit number.
- There are three common notation to show IP V4 address:
  - Binary
  - Decimal
  - Hexadecimal



IP Addressing

- Some errors in writing the IP address:
  - 111 . 56 . 045 . 78
    - There are no leading zeros in the decimal notation.
  - 221 . 34 . 7 . 8. 20
    - There is only four octets in the IPv4 address.
  - 75. 45 . 301 . 14
    - The maximum value of an octet is 255.
  - 11100010. 23. 14. 67
    - You can’t mix binary and decimal notations

IP Address Operations

- Some operations can be applied to the 32-bit address:
  - Bitwise NOT operation
  - Bitwise OR operation
  - Bitwise AND operation

Bitwise NOT operation

Input	Output
0	1
1	0

Number	00010001	01111001	00001110	00100011
Complement	11101110	10000110	11110001	11011100

Bitwise OR operation

Input1	Input2	Output
0	0	0
0	1	1
1	0	1
1	1	1

Number 1	00010001	01111001	00001110	00100011
Number 2	11101110	10000110	00000001	11011100
Output	11111111	11111111	00001111	11111111

Bitwise AND operation

Input1	Input2	Output
0	0	0
0	1	0
1	0	0
1	1	1

Number 1	00010001	01111001	00001110	00100011
Number 2	11111111	11111111	10001100	00000000
Output	00010001	01111001	00001100	00000000